GENESIS OS / RELATIONSHIP SCORES

COMPANION SYSTEMS ENGINEERING GUIDEBOOK

Emotional Genome · Anger Hex · Drift Gate · E_GRIEF · ExsAnonymous · RS Game Ecosystem

Purpose: This document extends the master blueprint into a build-ready guide for the companion systems that make Genesis OS adaptive, stateful, and emotionally intelligent in production. It locks the missing bridge layer: governance rules, score influence caps, logging standards, workflows, service boundaries, and AI control logic.

Audience	Product, backend, mobile, AI, data, and QA teams
Status	Recommended production spec using current blueprint and workbook as the deterministic base, with added runtime governor rules
Source basis	Master Engineering Blueprint, scoring workbook v4.1, Emotional Genome Matrix, Anger Hex, Event Labeling Heuristics, DriftEligibility, Taxonomy Lock File, E_GRIEF, ExsAnonymous, and RS Game Ecosystem

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1. What this guidebook locks now

The existing master blueprint is strong at philosophy, scoring structure, and product vision. This companion guide locks the operational bridge so development can build the adaptive systems without guessing. The emphasis is deterministic governance first, AI interpretation second, and controlled runtime updates across all modules.

- A cross-system governor that decides what can update immediately, what can update weekly, and what cannot update without drift eligibility.
- A production-grade Emotional Genome model with weighted inputs from assessment, messaging, Sanctuary, grief, ExsAnonymous, and validated games.
- A concrete runtime specification for Anger Hex, Event Labeling, Shadow Activation, and DriftEligibility.
- A full development blueprint for ExsAnonymous, including services, data objects, privacy controls, moderation, and AI outputs.
- A split game architecture: Singles games operate under the user's RS Code identity; Couple games operate under the shared couple username/handle selected when the relationship is confirmed.
- Influence caps so grief, venting, games, and chat behavior enrich the engine without corrupting core compatibility or identity labels.

Non-negotiable runtime rule
Design rule: no companion module is allowed to directly overwrite a slow identity trait on its own. Every module produces evidence. The governor decides whether that evidence may affect fast state, medium traits, or slow identity.

2. System architecture and control philosophy

Genesis OS should be built as an event-driven emotional operating system. Everything begins as a typed event, is transformed into canonical tags, is rolled into state vectors, and only then becomes a score, forecast, recommendation, or AI narrative. This makes the system auditable and reduces model drift.

> Raw Input → Event Logger → Normalizer → Labeler/Classifiers → Feature Store → Deterministic Rule Engine → AI Interpretation Layer → Governor → UI/Score Updates

- Deterministic first: every visible score must trace back to stored fields and versioned formulas.
- AI as a specialist layer: AI may summarize, explain, moderate, and propose signals, but it does not directly mutate canonical profile IDs.
- Fast-state vs slow-identity separation: emotions, anger, grief, and momentary strain can update quickly; archetype, primary shadow, and baseline polarity must change slowly.
- Module isolation: ExsAnonymous, games, Sanctuary, messaging, and couple tools write to their own event streams, then feed shared rollups rather than writing directly into each other's tables.
- Version everything: rules_version, ai_version, taxonomy_version, scoring_version, and prompt_version must be logged on every derived analysis row.

Update lane	What lives here	Update rule
Fast state	emotion cell, anger band, grief tier, conversation heat, daily readiness drag/boost	May update instantly from validated events and expires/decays quickly.
Medium traits	cluster confidence, communication style confidence, regulation style confidence, pattern fingerprint	May update on rolling windows when confidence ≥ 0.75 and contradictions are not spiking.
Slow identity	primary archetype, secondary archetype, primary shadow type, baseline polarity/energy type	May update only when DriftEligibility passes and confidence ≥ 0.85 across multiple sources.

3. Canonical taxonomy and state contract

This section locks the exact enums and core state keys that all services must use. Internal names may never drift. Marketing copy may change, but stored IDs must remain canonical.

Domain	Canonical IDs	Implementation rule
user_move	WITHDRAW, PRESS_FOR_CLARITY, ESCALATE, APPEASE, REPAIR,	Single label only per event.

	JOKE_DEFLECT, UNKNOWN	
trigger_labels	REJECTION_CUE, AMBIGUITY, DISRESPECT_CUE, CONTROL_LOSS, UNFAIRNESS, SHAME_EXPOSURE, ABANDONMENT_PANIC, BETRAYAL	Multi-label allowed when score ≥ threshold.
shadow_type_id	MANIPULATOR, STONEWALLER, LOVE_BOMBER, CHAMELEON, SCOREKEEPER, SELF_SABOTEUR	Identity label changes only via slow drift gate.
archetype_id	ICEBOX, PHANTOM_SEEKER, REGULATED_GROWN_UP, ANALYZER, QUIET_EXIT, FIXER, ROMANTIC_IDEALIST, PERFORMER, SURVIVOR, TRANSLATOR	Never rename internally.
strategy_class	VALIDATE_THEN_ASK, SLOW_PACE, BOUNDARY_CLARITY, DIRECT_CLARIFICATION, REASSURE_AND_GROUND, MIRROR_AND_SUMMARIZE, PAUSE_AND_SPACE, LIGHTEN_WITH_HUMOR	Forecast engine outputs only canonical strategy classes.
contradiction_cluster_id	VULNERABILITY_LOOP, CONTROL_AVOIDANCE, INTIMACY_FEAR, FAIRNESS_DISSONANCE, SELF_IMAGE_SPLIT	Reflection clusters must map to canonical IDs.

Recommended exact state vector keys

- state_pre / state_post should contain: arousal_score, safety_score, clarity_score, repair_openness, attachment_threat, anger_band, anger_pressure, grief_sdi, contradiction_pressure, volatility_score, emotion_code, emotion_family, emotion_cell_id, social_orientation.
- All state vectors should also store source_module, confidence, rules_version, timestamp, and window_scope (event / session / 24h / 7d / 30d).
- Every service reading or writing state vectors must validate keys against the taxonomy contract before persisting.

4. Emotional Genome Engine

The Emotional Genome is the multi-layer blueprint that converts all emotional inputs into structured pattern intelligence. It should be treated as the stable profile memory of Genesis OS. The genome does not react to every mood swing; it accumulates evidence, forms clusters, and updates under governance rules.

4.1 Input streams

Stream	Examples of captured signals	Primary outputs

Assessment baseline	60-question answer bank, weighted answer rows, contradiction seeds	archetype vectors, shadow vectors, polarity baseline, readiness baseline
Messaging	trigger labels, user_move, silence gaps, repair attempts, response rhythm, anger events	communication health, rupture/repair trend, drift vectors
Sanctuary	stillness sessions, Whisper Window usage, emotional mirror prompts, ritual completions	emotion cells, regulation support, grief overlays
ExsAnonymous	private venting, circle participation, partner autopsy analysis	pattern fingerprints, grief load, shadow evidence, attachment loops
Games	scenario choices, cooperation behavior, honesty depth, prediction accuracy, repair after loss	validated soft tags, growth credits, bonding metrics
Couple mode	pulse checks, repair exercises, shared reflections, ritual completion	runtime forecast overlays, bonding index, couple drift signals

4.2 Genome build layers

- Layer 1 — raw streams: all module events arrive with canonical module_id, event_type, state_pre/state_post, payload_json, and confidence.
- Layer 2 — tags: events emit small pattern tags such as Silent_Protection, Hyper_Analysis, Grief_Marked, Scorekeeping, Future_Fantasy, or Self_Sabotage_Signal.
- Layer 3 — clusters: tags accumulate into clusters such as Withdrawal, Pursuit, Fixer, Analyzer, Grief Imprint, Retraction Shadow, or Fantasy Attachment.
- Layer 4 — vector scores: clusters push weighted values into the archetype vector, shadow vector, regulation vector, communication vector, and polarity balance.
- Layer 5 — governor writeback: the governor decides whether results remain fast-state only, become medium-trait evidence, or are eligible to update baseline identity.

4.3 Weighted merge model

SlowTraitCandidate = 0.60*Assessment + 0.15*MessagingRollup + 0.10*ExsRollup + 0.05*Sanctuary + 0.05*GriefHistory + 0.05*ValidatedGames

MediumTraitCandidate = 0.45*Assessment + 0.20*MessagingRollup + 0.15*ExsRollup + 0.10*Sanctuary + 0.10*ValidatedGames

FastState = 0.40*RecentMessaging + 0.25*Sanctuary + 0.20*ExsRecent + 0.15*GriefRecent

Recommended production rule: assessment should remain the anchor source for identity. Companion systems can enrich confidence and detect change, but not overpower the baseline without stable evidence.

4.4 Genome confidence model

GenomeConfidence = 0.45*SignalConsistency + 0.25*CrossSourceAgreement + 0.15*RecencyDensity + 0.15*(1 - ContradictionPressure)

- Confidence < 0.55: advisory only; keep as internal evidence.
- 0.55–0.74: create soft tags and show AI language as possibility, not certainty.
- 0.75–0.84: medium-trait eligible if the same pattern appears across at least two modules.
- ≥ 0.85: slow-trait candidate; may update baseline only if DriftEligibility passes.

4.5 Required genome outputs

- archetype_vector_json: normalized scores for all 10 archetypes
- shadow_vector_json: activation baselines for all 6 shadows
- cluster_vector_json: confidence per cluster
- communication_style_confidence, regulation_style_confidence, attachment_tone_confidence
- pattern_stability_score, contradiction_pressure, grief_imprint_score, anger_profile_baseline
- genome_write_lock_state: FAST_ONLY / MEDIUM_READY / DRIFT_ELIGIBLE / LOCKED

5. Anger Hex runtime system

Anger Hex converts raw anger behavior into trackable event data without reducing the user to a permanent label. It should run as a fast-state classifier. The engine measures anger pressure, direction, and relational cost. It should feed forecast and intervention logic immediately, but identity updates must still go through drift rules.

Band	Definition	Primary product response
R1 — Disruption (#FFB36A)	Small irritation; something broke flow or expectation.	No score penalty. Log as low-risk irritation and monitor repeat rate.
R2 — Violation (#FF894D)	Minor standard or boundary crossed; agitation rising.	Tag event, increment tension count, watch repetition.
R3 — Threat (#FF5A36)	Pressure is now personal or identity-linked; frustration.	Increase rupture risk and lower forecast confidence slightly.
R4 — Defense (#E43117)	Armor mode; boundary protection is active.	Offer pause/clarity strategies and flag repair sensitivity.
R5 — Fire (#B10F0F)	Outburst or explosive discharge.	Immediate rupture flag, larger forecast penalty, cooldown intervention.
R6 — Collapse (#5A0B0B)	Imploded anger; shut-down, numbness, disappearance.	Silence-gap watch, stonewall risk bump, PAUSE_AND_SPACE intervention.

5.1 Runtime scoring model

AngerPressure = clamp(0,100, 30*TriggerSeverity + 20*MetadataHostility + 20*MoveEscalation + 15*RepeatPattern + 15*ArousalDelta)

- TriggerSeverity is built from disrespect, unfairness, betrayal, control loss, and shame exposure labels.

- MetadataHostility uses all-caps ratio, profanity count, hostile punctuation, short-cold replies, and abrupt latency compression.
- MoveEscalation maps user_move to pressure: WITHDRAW=0.15, PRESS_FOR_CLARITY=0.30, APPEASE=0.20, JOKE_DEFLECT=0.10, REPAIR=-0.10, ESCALATE=0.60.
- If AngerPressure ≥ 70 and output direction is internalized/withdrawn, classify as R6 rather than R5.
- Store anger_direction as OUTWARD / INTERNAL / MIXED and anger_cost as NONE / MINOR / MODERATE / HIGH.

5.2 Anger influence rules

- R1–R2: no readiness change; log frequency only.
- R3: runtime forecast penalty up to -4 if repeated in 7 days.
- R4: runtime forecast penalty up to -8 and strategy recommendations must shift toward pause, direct clarification, or reassurance.
- R5: immediate rupture flag, runtime forecast penalty -12 to -20 depending on repair attempt within 24 hours.
- R6: stonewall/silence risk bump and repair latency watch; penalty similar to R5 if unresolved beyond 48 hours.
- Anger events may enrich shadow activation, especially for STONEWALLER, SCOREKEEPER, and SELF_SABOTEUR patterns.

6. Event labeler and messaging telemetry

The Event Labeler is the autopilot bootstrap service that turns free text into structured trigger labels, a single user_move, markers, and confidence. It should run on each message, call attempt, or meaningful silence-gap event. In v1 it is rule-based; later it can be assisted by learned classifiers.

6.1 Inputs and outputs

Input/Output	Fields	Notes
Inputs	text, prev_text, metadata, genome_snapshot, state_pre	metadata includes response latency, message length, punctuation, caps ratio, profanity count
Outputs	trigger_labels[], user_move, markers[], confidence	trigger labels are multi-label; user_move is single-label only
Persistence	MessageEvent, EventLabelRow, StateTransitionRow	store raw text separately from structured event rows for privacy control

6.2 Required preprocessing

- Normalize lowercase text and whitespace.
- Keep punctuation counts, repeated punctuation, and all-caps ratio.
- Count profanity, “always/never” language, and time pressure phrases such as “right now” or “answer me.”

- Bucket message length as short (<12 tokens), medium, or long (>80 tokens).
- Compute question density, cold-short reply indicator, and contradiction-callout indicator ("you said X then Y").

6.3 v1 rule seed examples

Trigger	Seed logic from current docs	Recommended extensions
REJECTION_CUE	High signal phrases like "you don't care," "ignored," "ghost," "I'm not important," plus short cold "fine."	Add abandonment context weighting when same phrase appears after silence gap > user baseline.
AMBIGUITY	Question density, "what do you mean," "which is it," contradiction callout.	Boost when prior message had mixed reassurance/withdrawal markers.
DISRESPECT_CUE	Insults, name-calling, "shut up," contempt, sarcasm markers with "you."	Add mockery and dismissive-laughter cues later.
CONTROL_LOSS	"Don't tell me what to do," "stop trying to control me."	Also trigger from heavy command clusters and pressure phrases.
UNFAIRNESS / SHAME_EXPOSURE / BETRAYAL	Should be added with explicit phrase sets and contradiction context.	Keep atomic; do not create new pseudo labels.

6.4 user_move classification

- WITHDRAW: silence, detachment, short cold endings, "I'm good," "whatever," disappearing after tension.
- PRESS_FOR_CLARITY: direct explanation seeking, repeated questions, demand for consistency.
- ESCALATE: attack, contempt, threat, repeated hostile pressure.
- APPEASE: over-apology, self-blame, collapse, "it's my fault," "forget it, I'm sorry."
- REPAIR: ownership, reassurance, concrete reconnection attempt, "I see how that landed, let's fix it."
- JOKE_DEFLECT: humor, sarcasm, meme-like minimization, or teasing used to avoid direct tension.

7. DriftEligibility and Shadow Activation

This is the governance layer that prevents identity chaos. Shadow activation should be dynamic and event-driven. Shadow identity labels should be slow and evidence-based. DriftEligibility decides when the system may move a slow trait; Shadow Activation decides what is lighting up right now.

7.1 DriftEligibility() gate

Eligible if: total_events ≥ 50 AND unique_days_covered ≥ 10 AND E20 ≤ 0.35 AND A_std50 ≤ 0.22 AND |A_slope50| ≤ 0.05 AND (C50 ≤ 0.55 OR contradiction improvement ≥ 0.10) AND H_norm ≤ 0.78

EligibilityConfidence = 0.25*(1-E20/0.35) + 0.20*(1-A_std50/0.22) + 0.20*(1-|A_slope50|/0.05) + 0.20*(1-C50/0.55) + 0.15*(1-H_norm/0.78)

- E20 = escalation rate over last 20 events.
- A_std50 = arousal volatility over last 50 events.
- A_slope50 = linear trend of arousal over last 50 events.
- C50 = mean contradiction severity over last 50 events.
- H_norm = normalized move entropy across WITHDRAW / PRESS_FOR_CLARITY / ESCALATE / APPEASE / REPAIR / JOKE_DEFLECT.
- Even when the boolean gate passes, baseline identity should only update when EligibilityConfidence ≥ 0.60 and GenomeConfidence ≥ 0.85.

7.2 Shadow activation model

$$\text{ShadowActivationDelta} = \text{BaseTrigger} + \text{MoveBonus} + \text{AngerBonus} + \text{RepetitionBonus} + \text{ContextBonus} - \text{RepairCredit}$$

$$\text{ShadowActivation14d} = \text{clamp}(0,100, \text{prior_score}*0.90 + \Sigma\ \text{recent_deltas})$$

- 0–24 = latent, 25–49 = flicker, 50–69 = active, 70–100 = dominant episode.
- Decay should happen daily or on each recompute. Lack of reinforcing events must lower runtime activation.
- The profile field primary_shadow_type_id changes only after multiple 30-day windows support the same direction and DriftEligibility passes.

Shadow	Primary event cues	Recommended event writebacks
MANIPULATOR	future-faking, selective reassurance, push-pull pacing, strategic ambiguity	future_fake_flag, reassurance_spike, ambiguity_after_intensity
STONEWALLER	withdrawal after tension, silence gaps, low repair follow-through, R6 collapse	stonewall_event, silence_gap_count, unresolved_48h_flag
LOVE_BOMBER	intensity spikes early, overpromising, dramatic exclusivity pacing	intensity_surge, exclusivity_acceleration, affection_spike
CHAMELEON	opinion mirroring, rapid self-position shifts, acceptance-driven adaptation	mirroring_flag, values_shift_flag, self_erasure_signal
SCOREKEEPER	receipt-keeping, fairness tallying, resentful memory stacking	receipt_language, fairness_charge, unresolved_grievance_count
SELF_SABOTEUR	pre-emptive exits, closeness panic, fight-picking near intimacy	closeness_exit, rupture_before_depth, intimacy_fear_bump

8. E_GRIEF spiritual depletion engine

E_GRIEF is the grief-state engine that tells the platform when the user is not simply sad, but spiritually depleted. It is a fast-state protection layer and a product governor. Its job is not to diagnose; its job is to adapt pacing, prompts, social exposure, and Sanctuary behavior so the app becomes safer when grief is high.

8.1 Core score and tiers

```
SDI_raw = SessionScore + LanguageScore + SelahScore + UsagePatternScore
SDI = min(100, SDI_raw)
```

- 0–20 Stable
- 21–40 Tender
- 41–70 Active Grief
- 71–100 Deep Grief / Spirit Heavy

8.2 Recommended runtime behavior

Tier	Product behavior	Score/visibility rule
Stable	Normal Sanctuary, Selah optional, full app access.	No grief drag.
Tender	Softer AI tone, gentle Selah prompts, lower notification pressure.	Readiness drag up to -2; do not alter compatibility.
Active Grief	Auto-Selah after heavy sessions, no pushy social prompts, Whisper Window becomes expression-first.	Readiness drag up to -5; match surfacing throttle.
Deep Grief	Sanctuary defaults to stillness, notifications reduced, ExsAnonymous and Last Light become holding spaces.	Readiness drag up to -8; freeze growth pressure and avoid urgency messaging.

8.3 Decay and recovery

```
SDI_t+1 = max(0, SDI_t - DailyDecay + NewGriefLoad - RecoveryCredit)
```

- Suggested DailyDecay: 4 points when no grief-triggering event occurs in 24h.
- Suggested RecoveryCredit: 1–3 points for completed grounding or ritual actions, capped to prevent gaming.
- E_GRIEF should never directly rewrite archetype or primary shadow; it affects fast state, readiness drag, and feature exposure only.

9. ExsAnonymous development blueprint

ExsAnonymous is the emotional detox system inside Sanctuary. It needs to be built as a first-class subsystem, not a text box with a chatbot attached. The module must support three modes — Private, Circle, and Partner — while maintaining anonymity, clean event output, and safe AI behavior.

Privacy rule

Core rule: ExsAnonymous captures emotional patterns, not public identity. No names should be stored in structured outputs. Raw text should be redacted or encrypted and retained only according to the chosen privacy policy.

9.1 Shared technical architecture

- Exs Session Service: creates session_id, mode, user_id, consent_state, and module context.
- Transcript Service: stores text/voice turns, redacts names and direct identifiers, and generates a sanitized transcript version.
- Exs Analyzer: sends sanitized transcript plus state context to AI and rule engine.
- Pattern Fingerprint Writer: stores structured outputs such as attachment loops, archetype evidence, shadow evidence, grief load, contradiction hints, and ritual recommendations.
- Archive Writer: creates optional user-facing summaries for Archive of Self, with separate visibility rules from internal feature vectors.
- Governor Hook: converts Exs evidence into soft tags, runtime state updates, or drift candidates using confidence thresholds.

9.2 Data objects

Object	Core fields	Notes
exs_session	session_id, user_id, mode, started_at, ended_at, duration_sec, created_from_module, consent_state	Top-level session container.
exs_turn	turn_id, session_id, speaker_type, raw_text_encrypted, sanitized_text, voice_flag, created_at	speaker_type = USER / AI / MODERATOR.
exs_analysis	session_id, ai_version, rules_version, theme_json, pattern_fingerprint_json, grief_score, contradiction_hints_json, ritual_recommendation_json, confidence_json	Primary structured analysis row.
exs_archive_entry	archive_entry_id, user_id, session_id, mode, title, summary_text, visibility_state, created_at	User-facing memory object.
circle_room	room_id, room_topic_id, status, max_size, ai_moderation_level, created_at	Used only in Circle Mode.
circle_participant	room_id, anon_seat_id, user_id, mode_state, joined_at, left_at	RS identity hidden; use anonymous seat.

partner_autopsy	autopsy_id, user_id, target_type, intensity_rating, pattern_map_json, closure_stage, readiness_effect, created_at	Used in Partner Mode.

9.3 Private Mode workflow

- User enters Private Mode from Sanctuary.
- Session service starts exs_session(mode=PRIVATE) and takes a state snapshot.
- User types or speaks; transcript service stores encrypted raw text and a sanitized version.
- Labeler + analyzer identify emotional themes, loops, grief signals, shadow evidence, and contradiction hints.
- AI returns: emotional translation, stabilizing response, optional ritual, and confidence-rated pattern outputs.
- Governor writes soft tags and runtime state changes only; no slow-trait mutation from one session.
- Archive writer creates a private entry if the user opts in or the product design defaults it on.

9.4 Circle Mode workflow

- Rooms are topic-based and anonymous: Breakup Recovery, Ex Guilt & Self-Blame, Situationship Spiral, Night-Time Anxiety, and similar.
- Each participant receives an anonymous seat ID; singles should still not expose their RS Code here unless explicitly desired in a separate product decision.
- AI moderation maintains tone stability, blocks conflict/shaming, summarizes room themes, and detects macro emotional patterns.
- Circle outputs should be collective and anonymized. Do not let Circle Mode directly change slow identity. It is stronger at macro trend detection than precise trait reassignment.
- Store only the minimum necessary group analytics: room_topic_id, share_count, listen_count, intervention_count, conflict_block_count, room_theme_vector.

9.5 Partner Mode workflow

- User chooses target type: ex, almost-relationship, recent breakup, or current confusion.
- Session begins with a guided intake: intensity, duration, unresolved questions, strongest trigger, biggest loop.
- AI leads a structured autopsy across attraction pattern, polarity role, conflict loop, abandonment imprint, and reenactment risk.
- Outputs include a cycle map, likely loop summary, closure ritual, readiness calibration suggestion, and pattern tags stored anonymously.
- Use cautious language in prompts and outputs. This mode should offer pattern probabilities, not pseudo-diagnostic certainty about the other person.

9.6 ExsAnonymous → engine writeback rules

$$ExsPatternConfidence = 0.40*LanguageEvidence + 0.25*RepetitionAcrossSessions + 0.20*CrossSourceAgreement + 0.15*RecencyDensity$$

- Single-session outputs below 0.55 remain session-scoped only.
- 0.55–0.74 may create soft tags and ritual suggestions.
- 0.75–0.84 may update medium-trait evidence if messaging or Sanctuary agrees.
- ≥ 0.85 may participate in slow-trait updates, but only if DriftEligibility passes.

9.7 Moderation and safety requirements

- PII scrubbing: names, phone numbers, addresses, employer details, and social handles should be redacted from structured outputs.
- Conflict prevention: Circle Mode must block harassment, pile-ons, threats, and abuse; log moderation actions with rule IDs.
- Crisis protocol: if the safety model detects acute self-harm risk or high-risk abuse content, route the experience to a safety flow instead of standard reflective flow.
- Output caution: use “likely pattern,” “possible loop,” and “what this may be reflecting” rather than courtroom language.

9.8 Setup checklist for development

- Create exs_session, exs_turn, exs_analysis, exs_archive_entry, circle_room, circle_participant, and partner_autopsy tables.
- Build PII redaction before AI analysis; never send raw identifying text to the analysis layer if not required.
- Implement a reusable session side panel showing state_pre, state_post, confidence, and interventions for QA/admin review.
- Version prompts and rules separately so false positives can be traced to either prompt drift or rule drift.
- Add analytics for completion rate, session depth, ritual acceptance rate, grief tier distribution, and moderation events.

10. RS Game Ecosystem blueprint

The game layer must be fun, sticky, and measurable — but it should not cheaply hijack core matching. The clean implementation is to split the game ecosystem into two namespaces: Singles Games and Couple Games. Singles use the RS Code identity. Couple games use the couple username/handle selected when the relationship is confirmed.

10.1 Namespace split

Game namespace	Displayed identity	Primary engine purpose
Singles Games	RS Code	Discovery, low-pressure connection, self-awareness, validated micro-signals
Couple Games	Couple username / selected couple	Bonding, repair practice, joint ritual,

	handle	couple retention, shared telemetry
Friends / social lounge	RS Code or optional social alias, depending product decision	Community DAU, non-romantic retention, broadening platform use

10.2 Singles game families

- Solo self-discovery: Green Flag Challenge, Shadow Spotting, Polarity Pulses, Emotional Trivia, Spark Scenarios.
- Pair discovery for singles: Blind Guess Bomb, Connection Roulette, Spark Test, Quick Draw, Speed Trivia.
- Telemetry objective: reveal cooperation style, honesty depth, curiosity, emotional flexibility, response under pressure, and willingness to repair after a miss.

10.3 Couple game families

- Bonding and growth: The Bomb Game, Vulnerability Rounds, Team Relay Challenges, Harmony Match, 36 Emotional Questions.
- Telemetry objective: predictability, shared understanding, repair effort, emotional honesty, support style, and collaboration under playful stress.
- Couple games should update bonding metrics and runtime forecast overlays, not baseline compatibility.

10.4 Required game telemetry

Metric	Singles meaning	Couples meaning
completion_rate	Does the user finish reflective or pair-play experiences?	Does the couple complete activities together?
cooperation_score	How collaborative is the user under low stakes?	How well do partners coordinate, listen, and recover from mistakes?
repair_after_loss	Does the user become dismissive or graceful after losing/missing?	Does the couple repair quickly after tension in play?
truth_depth_index	Does the user choose surface answers or honest depth when invited?	How often do both partners engage meaningfully in vulnerability rounds?
prediction_accuracy	Mostly n/a outside Spark Test.	Measures mutual knowing in Harmony Match and similar games.
pressure_behavior	Tracks joking, deflection, shutdown, escalation under timed pressure.	Useful for live couple-state overlays, not slow identity updates.

10.5 Score-protection rules for games

- Singles games may create soft evidence and growth points, but should not swing public compatibility directly.
- Validated single-player reflective games may contribute up to +3 readiness points per 30 days if paired with completion and consistency.

- Multiplayer singles games may enrich communication and pressure_behavior tags, but their contribution should be capped until supported by messaging or assessment evidence.
- Couple games should feed a separate Bonding Index and runtime forecast overlay; they should not rewrite the original compatibility score that came from the assessment engine.
- Tournament behavior, poker skill, and party-game skill should be classified as entertainment metrics only unless the game was intentionally designed as a psychological signal source.

10.6 Recommended couple metric

```
BondingIndex = clamp(0,100, 0.25*PulseCheckTrend + 0.20*RepairCompletion +
0.20*CoupleGameCollaboration + 0.15*SharedSanctuary + 0.10*MilestoneReflections +
0.10*StreakStability)
```

BondingIndex is a runtime couple-health metric, not a substitute for compatibility. It powers Symphonic Match evolution, couple leaderboards, and supportive prompts.

11. AI component architecture and adjustment rules

AI should be treated as a modular runtime service layer with clearly scoped jobs. It reads raw content and structured state, but it must always output structured proposals that the deterministic governor can approve, cap, or reject.

AI service	What it reads	What it may write
Emotion Analyzer	sanitized text, metadata, state_pre, taxonomy contract	theme_json, pattern_json, confidence, candidate tags
Moderator	Circle/Community events, severity/risk flags, recent room context	block/allow actions, moderation reason, de-escalation prompt
Reflection Composer	structured scores, state deltas, selected module context	user-facing summaries, rituals, explanations, coping prompts
Pair Explainer	compatibility components, forecast components, interaction rules	interpretable pair summary and strategy suggestions
Governor Assistant	module evidence packets and current write-lock state	recommended writes only; final writes remain rule engine controlled

11.1 What AI must not do

- AI must not directly overwrite archetype_id, primary_shadow_type_id, or baseline polarity scores.
- AI must not invent taxonomy labels outside the canonical contract.
- AI must not treat one vent session as clinical truth.
- AI must not surface hidden internal risk fields publicly in a shaming way.
- AI must not manipulate match rank or compatibility for monetization.

11.2 Recommended adjustment protocol

$$\text{AIProposal} \rightarrow \text{RuleEngineValidation} \rightarrow \text{InfluenceCapCheck} \rightarrow \text{GovernorApproval} \rightarrow \text{Writeback}$$

- Every AI proposal must include proposal_type, target_metric, suggested_delta, confidence, supporting_evidence, and expiry window.
- Rule engine validates taxonomy, source permissions, and lane permissions.
- InfluenceCapCheck enforces module-specific caps so no single subsystem can distort the profile.
- GovernorApproval decides whether the writeback is event-scoped, session-scoped, medium-trait, or deferred until drift reevaluation.

11.3 Suggested AI tuning dimensions

- Tone mode: spiritual / calm neutral / clinical-light. The core engine remains the same; only narration changes.
- Safety mode: ordinary, elevated, crisis-safe.
- Relationship context: single discovery, active conversation, coupled, grief support, ex-release.
- Output strictness: interpretive, evidence-heavy, or audit-facing.

12. Scoring, metrics, and influence caps

This section consolidates the most important production formulas and the recommended caps for companion systems. Anything below marked as recommended is a pre-user-data starting value and should be recalibrated after enough observed outcomes are collected.

Metric	Formula / rule	Recommended cap or note
Readiness (0–100)	Assessment baseline ± regulation/growth/stability/grief/runtime calibration	Keep companion module adjustments modest; assessment remains anchor.
Compatibility (350–850)	Use existing workbook formula based on B, A, S_risk, P, Astro, Num, IntegrityPenalty	Do not let games or grief overwrite this core score.
Forecast (0–100)	Base forecast from regulation, repair, communication, commitment pace, pattern stability, shadow risk	Runtime overlays allowed for anger, grief, and couple behavior.
PatternStability	Existing workbook metric or expanded stability composite	Good candidate for drift/forecast confidence.
SDI / grief state	Fast-state protection score	Should gate pacing, not identity.
ShadowActivation14d	Rolling 14-day score per shadow	Fast-to-medium evidence only unless drift eligible.
BondingIndex	Couple runtime health metric	Separate from compatibility.
GameGrowthBonus	Validated solo growth activities only	Cap at +3 readiness / 30d.
ExsCalibration	Confidence-rated readiness or	Cap at ±5 readiness / 14d.

	pattern calibration	
RuntimeForecastOverlay	anger + grief + repair + couple behavior	Cap to keep forecast interpretable.

12.1 Existing compatibility core from workbook

CompatibilityScore = clamp(350,850, 350 + 500*(0.30*B + 0.22*A + 0.18*(1-S_risk) + 0.12*P + 0.08*Astro + 0.05*Num) - IntegrityPenalty - ClusterPenaltyPair + InteractionCompatModifier)

12.2 Runtime forecast overlay recommendation

ForecastRuntime = clamp(0,100, BaseForecast - AngerPenalty - GriefIsolationPenalty - SilencePenalty + RepairBonus + CoupleGrowthBonus + InteractionForecastModifier)

- BaseForecast comes from the deterministic formula already in the blueprint/workbook.
- AngerPenalty should mainly activate on R4–R6 events.
- GriefIsolationPenalty should activate only for higher SDI tiers and should be modest.
- RepairBonus requires real repair evidence, not optimistic language alone.

12.3 Influence cap matrix

Module	May affect	Recommended cap
Messaging	forecast, drift, shadow activation, communication health	High; this is primary live relational evidence.
Sanctuary	fast state, readiness support, grief state	Medium; should support, not redefine identity.
ExsAnonymous	readiness calibration, medium traits, grief state	Moderate; cap readiness drift to ±5/14d.
Games — singles	soft tags, growth points, communication hints	Low-to-moderate; cap +3 readiness/30d and no direct compatibility rewrite.
Games — couples	BondingIndex, runtime forecast, Symphonic evolution	Moderate; no baseline compatibility mutation.
E_GRIEF	feature exposure, AI tone, readiness drag	High on pacing; zero authority on slow identity.
AI-only inference	narrative and recommendations	Zero authority unless validated by rule engine.

13. Data model and logging schema

A clean event model is the moat. Every subsystem should write into a shared event contract so analytics, drift computation, and model recalibration remain possible across time.

13.1 Common event envelope

Field	Purpose	Example
event_id	Global unique ID	evt_01H...
user_id / pair_id / relationship_id	Scope of event	user for solo events, relationship for couple events
module_name	Origin module	messaging, sanctuary, exs_private, exs_circle, game_singles, game_couples
event_type	Typed action	message_sent, grief_recompute, circle_share, couple_game_complete
state_pre / state_post	Before/after state vector	JSON with exact keys
payload_json	Module-specific fields	game results, text hashes, ritual metadata
derived_labels_json	trigger labels, user_move, emotion code, anger band	canonical IDs only
confidence_json	rule and AI confidence by feature	label_confidence, genome_confidence, grief_confidence
taxonomy_version / rules_version / ai_version	Audit trail	v1.0 / rules_2026_03 / ai_reflect_03
created_at, window_scope	Time and aggregation scope	event / session / 24h / 30d

13.2 Key tables to add or strengthen

- user_profile_core, user_profile_runtime, genome_snapshot, genome_rollup_7d, genome_rollup_30d
- message_event, event_label_row, conversation_rollup_daily, pair_rollup_30d
- anger_event, grief_profile, grief_event_log, sanctuary_session, sanctuary_intervention_log
- exs_session, exs_turn, exs_analysis, exs_archive_entry, circle_room, circle_participant, partner_autopsy
- game_session_singles, game_session_couples, game_action_event, game_metric_rollup_daily
- relationship_container, couple_profile, couple_pulse_check, couple_exercise_completion, bonding_rollup
- governor_decision_log for all medium/slow-trait write decisions

13.3 Special identity fields for game split

- Singles namespace: rs_code is the primary displayed identity in singles games and discovery spaces.
- Couple namespace: add couple_handle (the selected username when becoming a couple) and use that as the displayed identity in couple games, couple leaderboards, and Symphonic spaces.
- Do not leak the couple handle into singles matchmaking surfaces unless explicitly designed.

14. Workflow catalog

Below are the critical runtime workflows that development should implement. Each workflow should emit events, run deterministic processing, and only then call AI if that workflow requires interpretation or narration.

14.1 Onboarding and assessment completion

- Create user profile and assign RS Code.
- Store assessment answers using answer bank row schema with score_weight, compatibility_weight, archetype/shadow mappings, polarity tags, energy tags, cluster tags, integrity flags, drift_relevance, and forecast_relevance.
- Compute readiness, baseline vectors, contradiction clusters, communication/regulation/love summaries, and initial compatibility ingredients.
- Write genome_snapshot(version=onboarding) and mark slow traits as locked pending live evidence.

14.2 Message event processing

- Store raw message.
- Run labeler and state estimator.
- Classify trigger_labels, user_move, anger band, and state delta.
- Update conversation rollups and shadow activation deltas.
- Recompute runtime forecast overlays if relationship or active conversation exists.
- If thresholds are crossed, surface strategy_class guidance or Sanctuary reflection prompt.

14.3 ExsAnonymous session processing

- Create exs_session and store transcript turns.
- Sanitize content, run AI analysis, and store structured outputs.
- Update grief signals, pattern fingerprints, and readiness calibration candidates.
- Governor decides session-only vs medium-trait evidence.
- Write optional Archive entry and log intervention acceptance.

14.4 Singles game session processing

- Game session starts under RS Code identity.
- Log every meaningful game action with timer state, response choice, cooperation markers, and repair behavior.
- On completion, compute game-specific metrics, then convert them into soft evidence only.
- If the game is reflective/validated, award growth credit and update low-risk soft tags within cap limits.

14.5 Couple game session processing

- Game session starts under couple_handle identity within relationship container.
- Log collaboration, support, frustration recovery, prediction accuracy, and completion state.

- Update BondingIndex and runtime forecast overlay only.
- Trigger Symphonic Match evolution when milestone thresholds are met.

14.6 Nightly / weekly jobs

- Nightly: recompute SDI, anger rollups, pair rollups, soft evidence, and decays.
- Weekly: recompute genome confidence, medium-trait candidates, and pattern stability.
- Biweekly or monthly: run DriftEligibility and consider slow-trait updates.
- Model QA: sample governor decisions and false positives for review.

15. Service/API map

A monolith is acceptable at first, but the route surface should already reflect the logical service boundaries so the platform can split into microservices later without rewriting the product contract.

Service area	Representative endpoints/jobs	Notes
Assessment / scoring	POST /assessment/submit, POST /score/recompute, GET /profile/blueprint	Backed by answer bank and scoring workbook logic.
Messaging intelligence	POST /messages/send, POST /labeler/extract, POST /state/recompute	Runs event labeler and fast-state updates.
Sanctuary / grief	POST /sanctuary/session/start, POST /grief/recompute, POST /ritual/complete	Owns E_GRIEF and intervention logs.
ExsAnonymous	POST /exs/private/start, POST /exs/circle/share, POST /exs/partner/start, POST /exs/analyze	Must use sanitized transcript pipeline.
Games — singles	POST /games/singles/session/start, POST /games/singles/action, POST /games/singles/complete	Displayed identity = RS Code.
Games — couples	POST /games/couples/session/start, POST /games/couples/action, POST /games/couples/complete	Displayed identity = couple_handle.
Governor / drift	POST /governor/evaluate, POST /drift/check, nightly recompute jobs	Owns medium/slow write approvals.
AI runtime	POST /ai/analyze, POST /ai/moderate, POST /ai/compose	Should return structured proposals, not direct database writes.

16. Analytics, QA, and rollout plan

These modules are powerful enough to create false certainty if shipped without instrumentation. QA is not optional here. The system needs both product QA and behavioral QA.

16.1 Minimum dashboards

- Taxonomy coverage: percent of events mapping cleanly into canonical labels.
- Label confidence distribution by trigger and user_move.
- Shadow activation false-positive review queue.
- SDI distribution, tier dwell time, and recovery slope by module path.
- ExsAnonymous completion rate, ritual acceptance rate, archive save rate, moderation events.
- Singles game telemetry vs real messaging outcomes to validate whether games are generating useful signal.
- Couple BondingIndex vs actual couple retention and repair completion.
- Forecast calibration dashboard: predicted stability bands vs observed outcomes.

16.2 QA checklist

- Verify that every module writes valid taxonomy IDs only.
- Verify that AI cannot directly mutate locked profile fields.
- Verify that grief tier changes alter pacing and UI behavior but do not corrupt identity.
- Verify that games never rewrite compatibility score.
- Verify that drift updates do not occur when eligibility gate fails.
- Red-team privacy flows in ExsAnonymous and Circle Mode.
- Run regression suites whenever rule thresholds or answer bank mappings change.

16.3 Recommended rollout order

- Phase 1: lock taxonomy, event envelope, scoring formulas, messaging labeler, and governor skeleton.
- Phase 2: ship E_GRIEF, Sanctuary writebacks, and basic anger/runtime forecast overlays.
- Phase 3: ship ExsAnonymous Private Mode first, then Partner Mode, then Circle Mode with moderation.
- Phase 4: ship Singles Games with soft-tag protection.
- Phase 5: ship Couple Mode, couple_handle identity, Couple Games, BondingIndex, and Symphonic evolution.
- Phase 6: recalibrate thresholds using live outcomes and begin learned model training with human-reviewed labels.

Final implementation note
Best build posture: treat this as a living operating manual. The answer-bank weights, rule thresholds, and learned classifiers should evolve, but the governance model should stay strict. That is what keeps the system from turning into poetic chaos with pretty dashboards.